

In this lecture, we will now consider the kinematics / statics of fingers
 The velocity of contact points and finger points are equal

chapter 3, velocity of tool frame $V_{st}^s = J_{st}^s(\theta) \cdot \dot{\theta}$ body velocity $V_{st}^b = Ad_{q_{st}}^{-1} V_{st}^s$

: instantaneous velocity $F_i: \begin{matrix} C_i & O & C_i \\ \leftarrow \text{finger} \rightarrow \end{matrix} F_i$: finger point, C_i : contact point S, F, C : frame

$\therefore V_{S_i F_i}^b = Ad_{q_{S_i F_i}}^{-1} J_{S_i F_i}^s(\theta_{F_i}) \dot{\theta}_{F_i}$, $J_{S_i F_i}^s$: Jacobian for F_i relative to S_i ,
 θ_{F_i} : vector of joint angles for finger i

(constraints ex) frictionless point contact:  $\cdot \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} V_{F_i C_i}^b = 0$
 $\Rightarrow$ in general contact with wrench basis B_{C_i} , $B_{C_i}^T \cdot V_{F_i C_i}^b = 0$

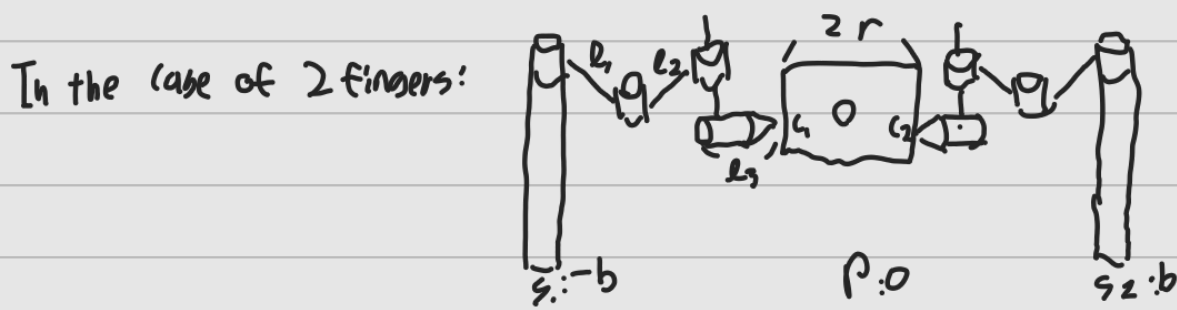
chapter 2: $V_{F_i C_i}^b = Ad_{q_{F_i C_i}}^{-1} V_{F_i P}^b + V_{P C_i}^b = -Ad_{q_{P C_i}}^{-1} Ad_{q_{P F_i}} V_{P F_i}^b + V_{P C_i}^b$
 $V_{P S_i}^b = 0 \Rightarrow V_{P F_i}^b = V_{S_i F_i}^b$, $V_{O C_i}^b = 0 \Rightarrow V_{P C_i}^b = Ad_{q_{O C_i}}^{-1} V_{P O}^b$
 $\therefore V_{F_i C_i}^b = -Ad_{q_{P C_i}}^{-1} Ad_{q_{P F_i}} V_{S_i F_i}^b + Ad_{q_{O C_i}}^{-1} V_{P O}^b$

(combine with constraint $\Rightarrow$) $B_{C_i}^T (Ad_{q_{P C_i}}^{-1} Ad_{q_{P F_i}} Ad_{q_{S_i F_i}}^{-1}) J_{S_i F_i}^s(\theta_{F_i}) \dot{\theta}_{F_i} = B_{C_i}^T Ad_{q_{O C_i}}^{-1} V_{P O}^b$
 $\Rightarrow B_{C_i}^T Ad_{q_{S_i C_i}}^{-1} J_{S_i F_i}^s(\theta_{F_i}) \dot{\theta}_{F_i} = G_i^T V_{P O}^b$ ($G_i = Ad_{q_{O C_i}}^{-1 T} B_{C_i}$)

hand Jacobian $J_h(\theta, x_0) = \text{diag}(B_{C_i}^T Ad_{q_{S_i C_i}}^{-1} J_{S_i F_i}^s(\theta_{F_i}))$, $x_0 = q_{P O}$
 $J_h(\theta, x_0) \dot{\theta} = G^T V_{P O}^b$ ($G = \begin{bmatrix} G_i \end{bmatrix}$): fundamental grasping constraint
 $\Rightarrow$ multifingered grasp is manipulable $\Leftrightarrow \mathcal{R}(G^T) \in \mathcal{R}(J_h(\theta, x_0))$ (range space)

$j \in N(J_h)$: internal motion

Section 5.2, object wrench $F_o = G f_c$, joint torque $\tau = J_h^T f_c$ (f_c : contact force)



Then $G = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$, $\theta_{C_i} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, $J_{S_i, C_i} = \begin{bmatrix} 0 & 1 & c_1 \sin \theta_1 & l_1 \cos \theta_1 + l_2 \cos \theta_1 \cos \theta_2 \\ 0 & 0 & c_1 \cos \theta_1 & -l_1 \sin \theta_1 + l_2 \sin \theta_1 \sin \theta_2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$\theta_{S_i, C_i}^{-1} = \theta_{\theta_{C_i}}^{-1} \theta_{p_0}^{-1} \theta_{C_i, p}^{-1} \Rightarrow \text{Ad}_{\theta_{S_i, C_i}}^{-1} = \text{Ad}_{\theta_{C_i}}^{-1} \text{Ad}_{\theta_{p_0}}^{-1} \text{Ad}_{\theta_{C_i, p}}^{-1}$
← constant →
↑ depends on λ_n

where $R_{p_0} = I$, $p_{p_0} = \begin{bmatrix} 0 \\ a \end{bmatrix} \Rightarrow R_{S_i, C_i} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$, $p_{S_i, C_i} = \begin{bmatrix} 0 \\ b-r \end{bmatrix}$, $R_{S_{K_i}} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $p_{S_{K_i}} = \begin{bmatrix} 0 \\ r \end{bmatrix}$

$\therefore \text{Ad}_{\theta_{S_{K_i}}}^{-1} = \begin{bmatrix} R_{S_{K_i}}^T & 0 & a & b-r \\ 0 & R_{S_{K_i}}^T & 0 & 0 \end{bmatrix}$, $\text{Ad}_{\theta_{S_i, C_i}}^{-1} = \begin{bmatrix} R_{S_i, C_i}^T & 0 & a & b-r \\ 0 & R_{S_i, C_i}^T & 0 & 0 \end{bmatrix}$, $J_n = \begin{bmatrix} J_{11} & 0 \\ 0 & J_{12} \end{bmatrix}$

$J_{11} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ r-b & l_1 r-b & l_1 r+l_2 r-b & 0 \\ 0 & l_1 c_1 & l_1 s_1+l_2 s_1 & 0 \\ 0 & l_1 s_1 & l_1 c_1+l_2 c_1 & 0 \end{bmatrix}$, $J_{12} = \begin{bmatrix} b-r & b-r+l_1 c_1 & b-r+l_1 c_1+l_2 c_1 & 0 \\ 0 & 0 & -l_1 s_1 & 0 \\ 0 & -l_1 c_1 & -l_1 s_1-l_2 s_1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

This case, grasp is not manipulable, $(C_i^T \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}) = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \notin R(J_n)$
 internal motions: $\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$

section 6: consider contact surfaces + rolling of objects against others
 considers smooth surfaces

local coordinate chart $c: U \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3, (u, v) \mapsto x$

$U \subset \mathbb{R}^2, \exists$ neighborhood $V \subset \mathbb{R}^3$, open $U \subset \mathbb{R}^2$, $c: U \rightarrow V$ is differentiable, homeomorphism
 $\forall q \in U, \frac{\partial c}{\partial u}(q): \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is injective $\Rightarrow S$ regular

$\mathcal{O}_p(U, V) \Rightarrow$ the tangent plane $= \text{span}(\frac{\partial c}{\partial u}, \frac{\partial c}{\partial v})$

$\frac{\partial c}{\partial u} \cdot \frac{\partial c}{\partial v} = 0 \Rightarrow$ orthogonal chart: locally exists for $\forall$ regular surface

First fundamental form of $\mathbb{R}^3$ $I_p: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}, \begin{bmatrix} c_u^T c_u & c_u^T c_v \\ c_v^T c_u & c_v^T c_v \end{bmatrix}$ ($p = (u, v)$)

positive definite $M_p: \mathbb{R}^2 \rightarrow \mathbb{R}^2, I_p = M_p \cdot M_p^T$, (c orthogonal $\Rightarrow M_p = \begin{bmatrix} \|c_u\| & 0 \\ 0 & \|c_v\| \end{bmatrix}$): metric tensor

Gauss map $N: S \rightarrow S^2, (u, v) \mapsto \frac{c_u \times c_v}{\|c_u \times c_v\|}$: normal unit of p

Second fundamental form: directional derivative of N

$$\Pi_p: \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}, \begin{bmatrix} C_u^T n_u & C_u^T n_v \\ C_v^T n_u & C_v^T n_v \end{bmatrix}, n_u = \frac{\partial n}{\partial u}, n_v = \frac{\partial n}{\partial v}$$

$$\text{Curvature tensor } K_p = M_p^{-T} \Pi_p M_p^{-T} = \begin{pmatrix} C_u^T n_u / \|C_u\|^2 & C_u^T n_v / \|C_u\| \|C_v\| \\ C_v^T n_u / \|C_u\| \|C_v\| & C_v^T n_v / \|C_v\|^2 \end{pmatrix}$$

(U, V) : orthogonal $\Rightarrow$ normalized Gauss frame $[1, 2] = \left[\frac{C_u}{\|C_u\|}, \frac{C_v}{\|C_v\|}, n \right]$

$$K_p: [1, 2]^T \begin{bmatrix} \frac{n_u}{\|C_u\|} & \frac{n_v}{\|C_v\|} \end{bmatrix}$$

$$\text{Torsion form } T: \mathbb{R}^2 \rightarrow \mathbb{R}, (u, v) \mapsto \begin{bmatrix} \frac{C_v^T C_{uv}}{\|C_u\|^2 \|C_v\|} & \frac{C_u^T C_{uv}}{\|C_u\| \|C_v\|^2} \end{bmatrix}$$

curve $p(t) \in S$, contact frame $C(t)$: Gauss frame of $p(t)$

if the frame of object O : fixed $\Rightarrow$ motion of C : $\mathcal{G}_{ac}(t) \in SE(3)$

$p(t)$ lies in a single coordinate chart $c: U \rightarrow \mathbb{R}^2$, $d\eta = C^{-1}(p(t))$

$$V_{ac}^b = \left(\begin{bmatrix} M_{ij}^a \\ 0 \end{bmatrix}, \begin{bmatrix} 0 & -T M_{ij}^a \\ T M_{ij}^a & 0 \\ \hline -(K M_{ij}^a)^T & 0 \end{bmatrix} \right) \quad \begin{array}{l} M: \text{metric tensor, } K: \text{curvature tensor} \\ T: \text{Torsion form of } (c, U) \end{array}$$

$$\text{ex) } c(u, v): \begin{pmatrix} 0 & \cos u \cos v \\ 0 & \cos u \sin v \\ 0 & \sin u \end{pmatrix}, \{ -\frac{\pi}{2} < u < \frac{\pi}{2}, -\pi < v < \pi \} \Rightarrow K = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, M = \begin{pmatrix} 0 & 0 \\ 0 & \cos u \end{pmatrix}, T = \begin{pmatrix} 0 & -\frac{\tan u}{\rho} \end{pmatrix}$$

$p_0(t) \in S_0, p_f(t) \in S_f$: point of contact, $S_0, S_f \Rightarrow (c_0, U_0), (c_f, U_f)$: charts

$\phi_0 = c_0^{-1}(p_0) \in U_0, \phi_f = c_f^{-1}(p_f) \in U_f$: local coordinate, C_0, C_f : orthogonal

ψ : angle between $\frac{\partial C_f}{\partial u_f}, \frac{\partial C_0}{\partial u_0}, \eta = (\phi_f, \phi_0, \psi)$: contact coordinate

$W = \{ \forall \mathcal{G}_{af} \} \subset SE(3), C_0, C_f$: coordinate frame coincide with normal Gauss frame at $p_0(t), p_f(t)$.

$L_0(t), L_f(t)$: coordinate frame relative to $O, F, L_0(t) = C_0, L_f(t) = C_f$

$V_{0|f}$: body translational velocity, $w_{0|f}$: body rotational velocity

if O, F contacts $\Rightarrow V \neq 0, \text{ Pure rolling contact: } V=0, w \neq 0, \text{ Pure sliding contact: } w=0, V \neq 0$

$$V_{b|f} = \begin{bmatrix} v \\ w \end{bmatrix} = \text{Ad}_{\phi_{f|f}}^{-1} V_{of} = \begin{bmatrix} R_{f|f}^T V_{of} - R_{f|f}^T \hat{p}_o w_{of} \\ R_{f|f} w_{of} \end{bmatrix}$$

orientation of x, y axis of C_f relative to x, y axis of C_o : $R_\psi = \begin{bmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{bmatrix}$

curvature of C at point of contact relative to x, y axis of C_f : $\tilde{\kappa}_o = R_\psi \kappa_o R_\psi$

$\kappa_f + \tilde{\kappa}_o$: relative curvature form

$$\left. \begin{aligned} \dot{\phi}_f &= M_f^{-1} (\kappa_f + \tilde{\kappa}_o)^{-1} \left(\begin{bmatrix} -w_y \\ w_x \end{bmatrix} - \tilde{\kappa}_o \begin{bmatrix} v_y \\ v_x \end{bmatrix} \right) \\ \dot{\phi}_o &= M_o^{-1} R_\psi (\kappa_f + \tilde{\kappa}_o)^{-1} \left(\begin{bmatrix} -w_y \\ w_x \end{bmatrix} + \kappa_f \begin{bmatrix} v_y \\ v_x \end{bmatrix} \right) \\ \dot{\psi} &= w_x + \tau_f M_f \dot{\phi}_f + \tau_o M_o \dot{\phi}_o \end{aligned} \right\} \begin{array}{l} \text{motion of the} \\ \text{contact coordinates} \\ \eta \end{array}$$

$$v_1 = 0$$

ex) sphere rolling on a plane: $\kappa_o = 0$, $\kappa_f = \frac{1}{r} I$, $M_o = I$, $M_f = \begin{bmatrix} 0 & 0 \\ 0 & m r^2 \end{bmatrix}$

$$\tau_o = 0, \tau_f = \begin{bmatrix} 0 \\ -\frac{\tan\psi}{r} \end{bmatrix}$$

$$\begin{bmatrix} \dot{v}_f \\ \dot{v}_f \\ \dot{v}_o \\ \dot{v}_o \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 0 \\ \sec\psi \\ -r \sin\psi \\ -r \cos\psi \\ -\tan\psi \end{bmatrix} w_x + \begin{bmatrix} -1 \\ 0 \\ -r \cos\psi \\ r \sin\psi \\ 0 \end{bmatrix} w_y$$

Grasping with rolling

$$J_h(\theta, x_0, \eta) \dot{\theta} = G^T(\eta) V_{po}^b, \quad \dot{\eta}_i = A_i(\eta_i) V_{of_i}^b \quad (\eta: \text{contact coordinates})$$

$B_{ci}^T V_{oci}^b \approx 0$: contact point cannot move in constrained directions

$\kappa_f + R_\psi \kappa_o R_\psi^T$ is full rank $\Rightarrow$ there is a smooth local bijection between η, θ_{of} ,

$\Rightarrow J_h(\theta, x_0) \dot{\theta} = G^T(\theta, x_0) V_{po}^b$, if not

$$\text{Ad}_{\phi_{s|c_i}}^{-1} = \text{Ad}_{\phi_{c_i}}^{-1} \text{Ad}_{\phi_{po}}^{-1} \text{Ad}_{\phi_{s|p}}^{-1}, \quad \phi_{oc_i} = \begin{bmatrix} \frac{\partial c_i}{\partial \theta_i} M_{o_i}^{-1} & n_{o_i} & c_i(\theta_i) \\ 0 & 0 & 1 \end{bmatrix}$$